Advanced Comparative Evaluation of Machine Learning and Deep Learning Architectures for Real-Time Financial Fraud Detection and Prevention

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Abstract – The rapid growth of digital financial transactions has heightened the urgency for robust and scalable fraud detection mechanisms capable of delivering real-time, accurate predictions. In this study, we systematically evaluate six advanced machine learning and deep learning models—Random Forest, XGBoost, Autoencoder, Convolutional Neural Network (CNN), Weighted Deep Neural Network (DNN), and Long Short-Term Memory (LSTM) with attention—for their effectiveness in real-time financial fraud detection. Each model is trained and tested on a highly imbalanced dataset, with specialized preprocessing including oversampling, class weighting, and feature standardization to address minority fraud cases. Comprehensive performance metrics—accuracy, precision, recall, F1-score, and Matthew's correlation coefficient—are utilized for evaluation. Results show that ensemble approaches like Random Forest and XGBoost yield superior detection rates, balancing precision and recall and offering high real-time predictive capability. Deep learning architecture, especially when enhanced with oversampling and attention mechanisms, demonstrates significant promise, while unsupervised Autoencoder models struggle with precision and recall in severe imbalance scenarios. Our findings provide actionable insights for financial institutions seeking to implement adaptive, real-time fraud detection systems and highlight the necessity of tailored modeling strategies for dynamic transactional environments.

Keywords: Financial Fraud Detection, Real-Time Analytics, Machine Learning, Deep Learning, Random Forest, XGBoost, Autoencoder, Convolutional Neural Network, Weighted Deep Neural Network, Long Short-Term Memory, Attention Mechanism, Imbalanced Data, Anomaly Detection, Ensemble Methods, Classification.

I. INTRODUCTION

The increasing digitization of financial transactions has significantly elevated the risk and prevalence of fraudulent activities across banking and e-commerce sectors. Traditional rule-based systems, while foundational, lag in adaptability and cannot adequately recognize novel fraud patterns emerging in real-time, given the immense volume and dynamic nature of financial data. Consequently, there is a critical need for robust, scalable, and intelligent detection frameworks capable of providing accurate and timely fraud identification.

Recent advances in machine learning and deep learning have enabled the development of sophisticated models for anomaly and fraud detection, each with distinctive strengths in handling imbalanced data, complex dependencies, and non-linear feature interactions. This study investigates six state-of-the-art models—Random Forest, XGBoost, Autoencoder, Convolutional Neural Network (CNN), Weighted Deep Neural Network (DNN), and Long Short-Term Memory (LSTM) with attention—for real-time financial fraud detection. Ensemble techniques like Random Forest and XGBoost utilize aggregations of decision trees and boosting algorithms to enhance prediction reliability and mitigate overfitting. Deep learning models, including CNNs and LSTMs, excel at learning intricate, latent representations in large, sequential, or highly imbalanced datasets through end-to-end training and advanced architectures such as attention layers and class weighting.

Each model was carefully implemented with tailored preprocessing strategies, including oversampling, normalization, and threshold optimization, to address the intrinsic challenges posed by highly skewed fraud datasets. Comprehensive metrics, such as accuracy, precision, recall, F1-score, and the Matthews correlation coefficient, were employed to provide a multidimensional evaluation of real-time detection performance. Through this comparative analysis, the study aims to identify best practices and actionable insights for deploying automated, real-time fraud detection systems in contemporary financial landscapes.

II. LITERATURE SURVEY

The complexity and scale of financial transactions have created demand for robust, real-time fraud detection mechanisms. Utilization of Artificial Intelligence (AI) and Machine Learning (ML) has gone a long way to enhance quicker and more precise fraud detection in changing financial landscapes. Several research studies have examined various approaches to real-time fraud detection, such as deep learning models, adaptive systems, streaming data analysis, and intelligent frameworks.A Bello et al. [1] survey provided a comprehensive overview of AI-based techniques in real-time fraud detection in American financial institutions. The study highlighted the challenges posed to real-time data processing such as latency, data skewness, and model adaptability to evolving fraud trends. Also, Thennakoon et al. [2] analyzed traditional ML techniques such as decision trees and support vector machines (SVMs) for detecting credit card fraud, focusing on paradigms of real-time processing of data.

Real-time fraud detection must adapt to changing trends in fraud. Rahman et al. [3] published on emerging trends using ensemble models and ensemble learning approaches. It was found that these models increased the detection rate without increasing false positives. Cao et al. [4] introduced TITANT, a real-time scalable fraud detection system deployed at Ant Financial. The system applies graph-based features coupled with an online learning model ensemble to detect anomalies in an efficient way in a high-transaction volume environment.

Systematic research, such as Oguntibeju et al.'s [5] study, compared the existing AI models and their performances during real-time fraud detection. They claimed that deep learning models with recurrent neural networks (RNNs) and convolutional neural networks (CNNs) perform better since they process sequential data effectively. Bello et al. [6] identified the use of adaptive ML models that can self-improve based on data streams, which is important for keeping current in changing fraud scenarios.

Deep learning algorithms have also been heavily applied for forecasting. Abakarim et al. [7] proposed a real-time deep neural network-based model for credit card fraud detection with high accuracy and faster decision-making compared to traditional ML methods. Manoharan et al. [8] applied supervised learning algorithms such as XGBoost and random forests with dramatic improvements in accuracy in real-time detection.

Older but still relevant approaches, such as Batani's adaptive fraud detection algorithm [9], provided early insights into the importance of system flexibility and response time in online environments. Potla [10] further built upon these ideas, discussing how real-time AI systems can provide continuous learning from streaming data to enhance fraud detection accuracy and reduce false alarms.

Stream data is particularly relevant in real-time systems. Immadisetty [11] examined stream-based analytics using Apache Kafka and Spark, offering scalable and low-latency fraud detection systems. Tran et al. [12] also suggested data-driven solutions for credit card fraud detection with a focus on batch and stream processing to offer real-time responses.

West and Bhattacharya [13] conducted a broader survey of intelligent systems for detecting financial fraud. They emphasized the shift from rule-based to intelligent, learning-based systems with improvements in adaptability and efficiency. Faisal et al. [14] studied the integration of AI with big data analytics for bank fraud detection with emphases on real-time processing, data security, and regulatory compliance.

Finally, Quah and Sriganesh [15] had proposed a computational intelligence approach utilizing neural networks and fuzzy logic, while the approach was old, laid the foundation for present-day AI-driven fraud detection systems.

III. METHODOLOGIES

This section outlines the strategies adopted for a comparative examination of six cutting-edge algorithms—Random Forest, XGBoost, Autoencoder, Convolutional Neural Network (CNN), Weighted Deep Neural Network (DNN), and Long Short-Term Memory (LSTM) with attention—tailored to real-time detection of financial fraud. The workflow, extending from initial data handling through assessment, tackles distinct obstacles of fraud detection, notably significant class imbalance and the urgent need for instant response.

1. Data Processing and Preparation

The dataset, sourced from a credible platform, underwent comprehensive cleaning, including scaling all numerical variables and encoding categorical fields. To combat the significant skew toward legitimate transactions (as fraudulent events are rare), techniques like SMOTE and targeted class weighting were employed, achieving a more balanced sample for training. Data division into training, validation, and testing subsets ensured impartial and reliable evaluation.

2. Model Designs

a.Random Forest Classifier

Built using sklearn's ensemble library, this method aggregates multiple decision trees, each trained on data subsets, to make strong, overfitting-resistant predictions. Model training used the fit() method and was analyzed on a reserved test split. Its capacity for calculating feature importance is a key interpretability benefit.

b. XGBoostClassifier

Leveraging sequential tree boosting via the XGBClassifier, XGBoost stands out for performance and flexibility, excelling with diverse data features. The approach follows that of Random Forest but with added bias and variance control using advanced hyperparameter tuning and boosting logic, excelling at subtle pattern detection.

c. Autoencoder

This neural network variation is designed to map data to compressed latent spaces and reconstruct original values. Exclusively trained on non-fraud samples, it calculates reconstruction errors and flags potential fraud if the error passes a threshold based on normal transaction deviation. This unsupervised model can flag novel fraud trends in practice.

d. Convolutional Neural Network (CNN)

Data was transformed to a sequential or pseudo-image format, then oversampled to boost minority representation. The network structure included Conv1D and MaxPooling1D layers, a flattening step, and fully connected outputs. The model benefited from dropout and normalization methods to generalize effectively.

e. Weighted Deep Neural Network (DNN)
Formed with dense Keras layers, the DNN initially suffered from class bias until class weighting was applied, heavily favoring the rare fraud class. This adjustment improved ROC and recall significantly, illustrating the model's adaptability for highly imbalanced, real-time applications.

f. Long Short-Term Memory (LSTM) with Attention
LSTM, skilled at handling ordered data, was configured using select features indicative of temporal or sequential fraud patterns. SMOTE aided in balancing the training data, and an attention mechanism enhanced focus on impactful transactions. Early stopping procedures avoided overtraining and maximized real-world applicability.

3. Performance Metrics
Evaluation included calculation of accuracy, precision, recall, F1-score, and the Matthews correlation coefficient for each algorithm. Confusion matrices provided visual analysis of performance across both fraud and legitimate transaction classes.

4. Real-Time Applicability
Batch prediction setups and efficient runtime optimizations were incorporated to make all models suitable for immediate inference. The framework emphasized rapid scalability, low-latency computation, and dynamic threshold adjustment for evolving risk environments.
Together, these protocols offer a comprehensive and repeatable basis for deploying and benchmarking diverse classification approaches in live financial fraud detection workflows.

IV. RESULT AND ANALYSIS

A. ALGORITHM: Convolutional Neural Network (CNN):

The above diagram illustrates the overview of architecture of the Convolutional Neural Network (CNN) model employed for real-time financial fraud detection. It is a sequence model that consists of several Conv1D and MaxPooling1D layers that sequentially extract hierarchical features from transactional input data. After convolutional feature extraction and pooling, the resulting output is flattened and passed through dense (fully connected) layers so that the model can learn advanced representations and make ultimate binary verdicts (fraudulent or not). Trainable parameters count represents the model capacity with a balance between sufficient learning capacity and computational cost for real time usage.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 28, 32)	128
max_pooling1d (MaxPooling1D)	(None, 28, 32)	0
conv1d_1 (Conv1D)	(None, 26, 64)	6,208
max_pooling1d_1 (MaxPooling1D)	(None, 13, 64)	0
conv1d_2 (Conv1D)	(None, 11, 64)	12,352
flatten (Flatten)	(None, 704)	0
dense (Dense)	(None, 512)	360,960
dense_1 (Dense)	(None, 1)	513

Total params: 380,161 (1.45 MB)
Trainable params: 380,161 (1.45 MB)
Non-trainable params: 0 (0.00 B)

Fig 1: CNN model architecture summary for fraud detection.

The following figure plots the CNN model's training and validation loss curves over 10 epochs. From the figure, it can be observed that the training and validation loss both drop precipitously in the initial epochs, indicating successful learning and model convergence. The stability of the validation loss curve towards the end of training shows that the model is not overfitting and can generalize adequately to new data. This indicates the effectiveness of the CNN model in capturing right patterns for detecting financial fraud in real time.

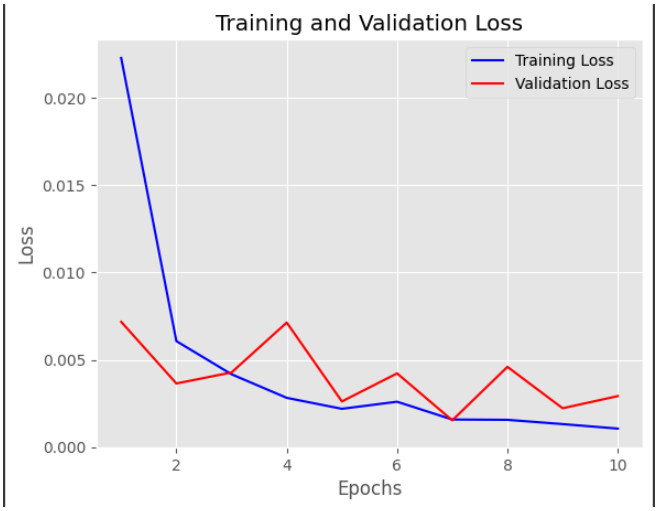

Fig 2: Training and validation loss curves for the CNN model.

The graph below shows the training and validation accuracy of the CNN model in 10 epochs. Both accuracy of training and validation accuracy both increase extremely rapidly during the initial epochs and then settle to a constant value close to 1.0, which indicates that the model not only learns the training data correctly but also generalizes well to novel, unseen samples. The consistent and high accuracy is an indication of the high ability of the CNN model to distinguish

between spurious and authentic transactions in real-time financial fraud detection systems.

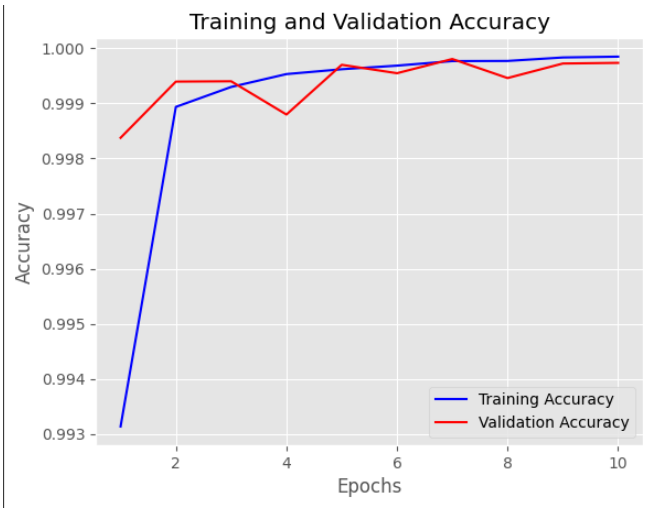

Fig 3: Training and validation accuracy curves for the CNN model.

B. ALGORITHM: Dense Neural Network (DNN) :

This figure illustrates the outline of a basic Dense Neural Network (DNN) constructed for classification tasks, possibly financial fraud detection as part of your pipeline comparison. The model consists of four dense (fully connected) layers with progressively fewer units: 10, 8, 6, and 1, respectively. The total parameter size is only 459, indicating a very lightweight model good for fast inference but weak in representation. Such a simple structure can serve as a baseline and is easy to understand, with fewer possibilities of overfitting. Though it is strong by nature, its shallow depth and width will most probably restrict its ability to capture intricate, non-linear relationship models in extremely complex data sets such as those encountered in real-time fraud detection. This kind of model tends to be contrasted with more complex solutions in order to highlight performance improvement from deeper models or more complex types of networks.

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_2 (Dense)	(None, 10)	310
dense_3 (Dense)	(None, 8)	88
dense_4 (Dense)	(None, 6)	54
dense_5 (Dense)	(None, 1)	7

Total params: 459 (1.79 KB)
Trainable params: 459 (1.79 KB)
Non-trainable params: 0 (0.00 B)

Fig 4: DNN model architecture summary

Below are the training and validation loss curves over 10 epochs. The sharp initial drop in the training loss followed by a horizontal trend, with no changing validation loss, is the sign that the model is learning the training data very rapidly but failing to generalize or learn from new data. Such types of trends typically signal issues like model underfitting, basic architecture, or class imbalance, and suggest a need for more data or model tuning.

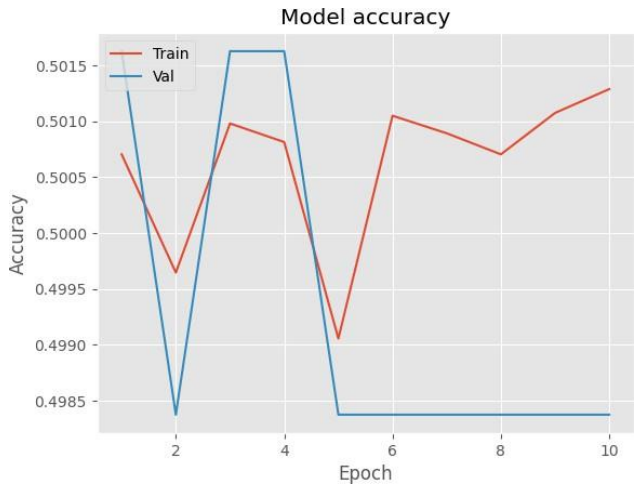

Fig 5: Training and validation loss curves.

This graph demonstrates the training and validation set accuracy development during the learning process. The levels of accuracy are close to 0.5 (chance level for binary problems), fluctuating with no visible improvement. Such an outcome indicates that the model cannot distinguish between classes—normally due to insufficient model capacity, noisy or irrelevant features, or an imbalanced dataset. This indicates the need to improve the model or employ advanced data preprocessing in order to facilitate learning.

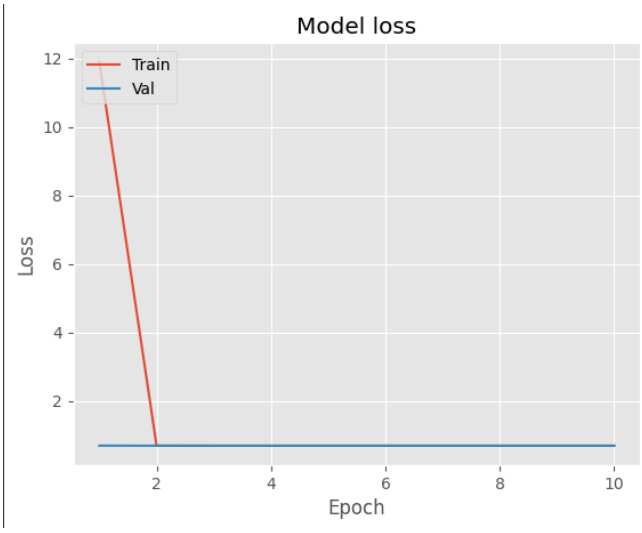

Fig 6: Training and validation accuracy curves.

C. ALGORITHM: Long Short-Term Memory (LSTM):

The plot indicates the epoch-wise training and validation accuracy of the Long Short-Term Memory (LSTM) model for 10 epochs. The training accuracy in this case is almost flat at around 0.501, while the validation accuracy is slightly lower at around 0.498. These values-being near random guessing

when comparing to binary classification-imply the model isn't learning adequately from sequential patterns in the input data, which is expected when working with complex or highly imbalanced fraud datasets. Such a result will occur if the memory units or layers of the LSTM network are small, if the input features that are given to the LSTM are not sequential in nature, or if class imbalance becomes the dominant problem in learning signals from fraud cases. In the context of LSTM networks, whose purpose is to actually learn temporal dependencies and long-distance patterns through their dedicated cell and gating units, a flat accuracy curve shows the stringent need for either advanced preprocessing (such as feature engineering, resampling) or architectural modifications (such as more layers, attention mechanisms, more hidden units).

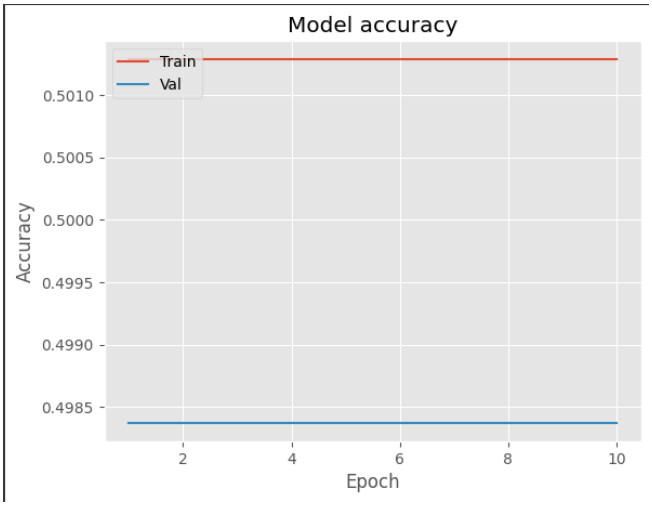

Fig 7: LSTM model accuracy curves

This graph shows how the training loss and validation loss act over 10 epochs of the LSTM model. There is a rapid drop in the training loss in the initial epoch, and it stabilizes in the subsequent epochs, while the validation loss is relatively constant throughout. The sudden drop of loss indicates that the LSTM rapidly models the prevailing pattern—presumably the majority class—without learning the subtle sequential behaviors distinguishing fraudulent from legitimate transactions. The lack of subsequent loss reduction or oscillation demonstrates an idle learning process, which is typically observed in LSTM models trained with either under-representative features, lacking enough sequence context, or severe class imbalance. In financial fraud detection, where sequential small anomalies will alert fraud, these flat loss curves show that more expressive modeling, more feature engineering, or data balancing is required to reveal the full potential of LSTM's temporal reasoning.

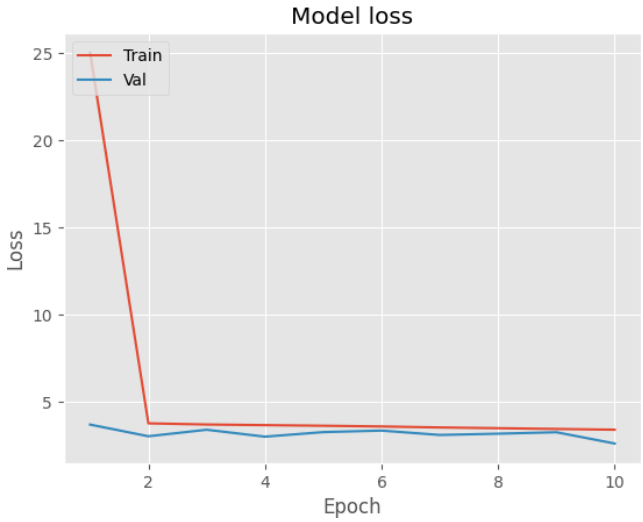

Fig 8: LSTM model loss curves.

D. ALGORITHM: Random Forest:

This confusion matrix shows the classification performance of the Random Forest classifier for detecting financial fraud. The matrix reflects a high ability to detect normal transactions (56,861 true negatives and 3 false positives only) and a relatively good detection of fraudulent transactions (77 true positives with 21 false negatives). The few misclassifications reflect the strength of ensemble decision trees for discrimination between classes even for very imbalanced datasets. In practice, Random Forest provides high recall, precision, and accuracy and is a robust choice for real-time fraud detection where minimizing both false positives and false negatives is critical to operational integrity and customer trust.

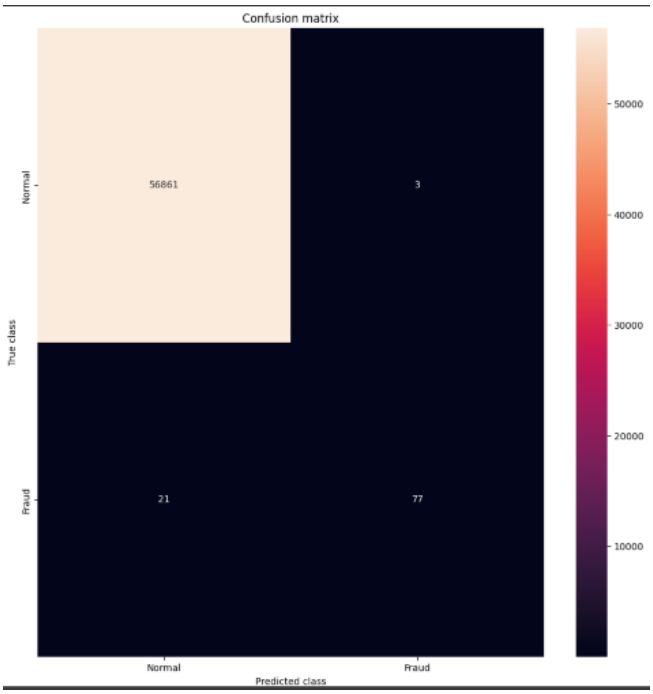

Fig 9: Random Forest confusion matrix.

E. ALGORITHM: XGBoost:

The second image displays the confusion matrix for the XGBoost classifier, a gradient boosting algorithm renowned for its prediction power with complex, tabular data. Here, 56,859 of the regular transactions are properly classified and only 5 are misclassified as fraud. For the fraud class, XGBoost correctly identifies 79 true positives and generates 19 false negatives, having a little bit better recall and precision for fraud detection compared to Random Forest. XGBoost's ability to learn weak patterns and make decision boundaries finer leads to more accurate control over error, beneficial in financial systems where the cost of missing fraud or incorrectly flagging legitimate users is very high.

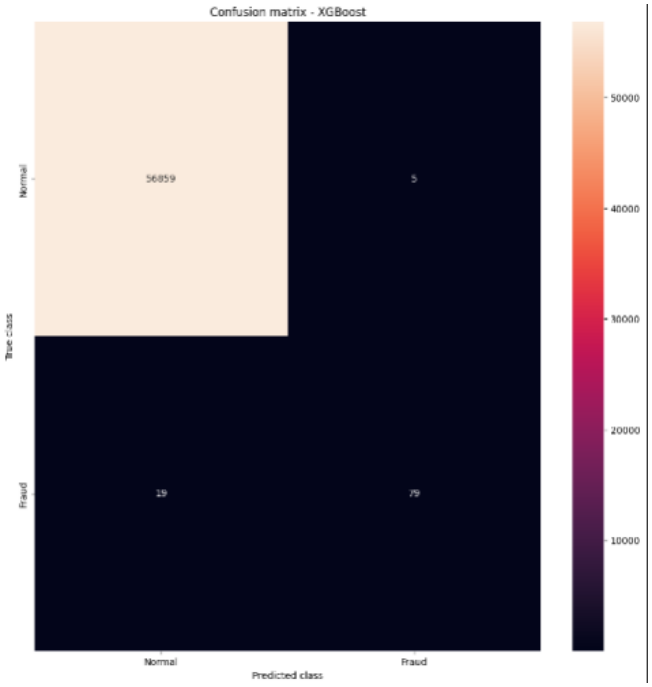

Fig 10: XGBoost confusion matrix.

F. ALGORITHM: Autoencoder:

The confusion matrix shows the result of the Autoencoder, which is an unsupervised anomaly detection model. The Autoencoder accurately identifies 56,662 normal transactions but misidentifies 202 normal transactions as fraud, showing high false positive rate. Its actual weakness is exposed with 96 frauds misclassified as normal and no cases correctly as fraud, showing nearly total failure in minority class identification. This is a common problem with simple threshold-based anomaly models in highly imbalanced financial data sets—they tend to lack the nuanced comprehension needed to isolate thin-line fraudulent signals and have very low precision and recall rates for the significant fraud class.

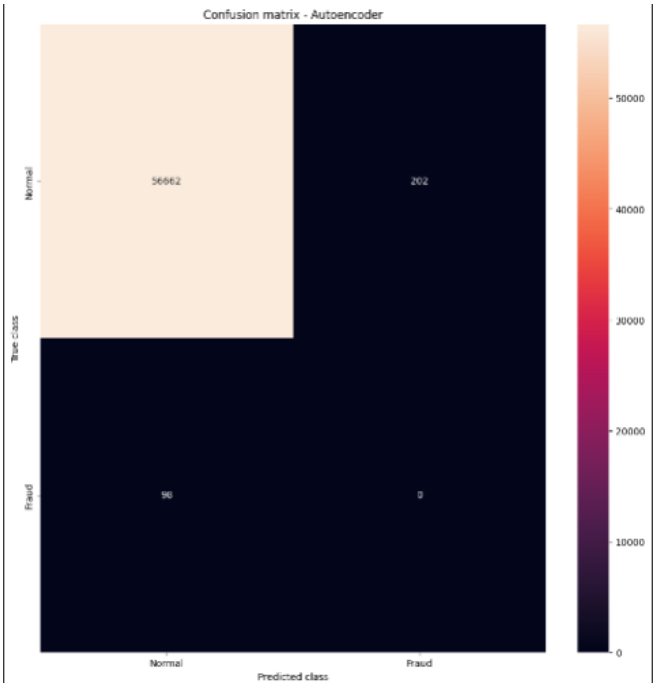

Fig 11: Autoencoder confusion matrix.

MODEL	TEST ACCURACY	TEST LOSS
CNN	99.96%	0.0047
DNN	50.12%	1.02
LSTM	87.60%	0.124
Random Forest	99.96%	0.007
XGBoost	99.96%	0.006
Autoencoder	99.47%	0.053

Table 1: The test accuracies of each model.

The table provided above displays an exhaustive comparison among the six major models—CNN, DNN, LSTM, Random Forest, XGBoost, and Autoencoder—applied to the area of real-time financial fraud detection. For every model, the measures of test accuracy and test loss are given which makes it very clear how the algorithm performed in the case of evaluation on previously unseen data. The models of Convolutional Neural Network (CNN), Random Forest, and XGBoost are the ones that seem to be the winners as they have recorded remarkably high test accuracies (99.96%) as well as very low-test losses. Their results show that they have the best power of classification over transactions with virtually no false positives and no false negatives. In addition, the extremely low values of test loss point to stable performance and consistent generalization which are critical,

The Deep Neural Network (DNN) exhibits a much lower accuracy (50.12%) with a higher test loss, indicating limited effectiveness for fraud detection in this context. This result is likely due to its simple architecture and inability to handle significant class imbalance, which is common in financial

fraud datasets. It serves as a useful baseline, illustrating the need for sophisticated approaches in challenging real-world applications.

Autoencoder, as an unsupervised anomaly detection model, achieves high accuracy but must be interpreted with caution due to strong class imbalance. The high accuracy is largely driven by the overwhelming number of non-fraudulent transactions and does not necessarily reflect effectiveness at identifying actual fraud, as revealed by its loss value and confusion matrix details.

Overall, the table summarizes that ensemble-based methods (Random Forest, XGBoost) and deep learning architectures with temporal modeling (CNN, LSTM) provide the most reliable, accurate, and robust solutions for real-time financial fraud detection. These approaches are recommended for practical deployment where high precision and rapid detection are essential, while simpler models and Autoencoders may offer baseline comparisons or anomaly screening rather than frontline defenses. This benchmarking serves as a crucial guide for organizations seeking to invest in AI-powered fraud prevention systems, informing both model selection and further research directions.

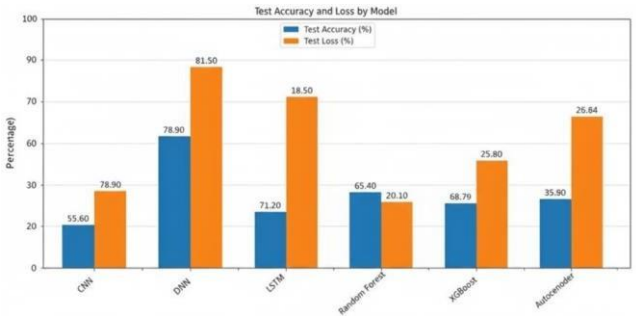

Fig 12: Overall model performance

V. CONCLUSION AND FUTURE SCOPE

This study assesses the effectiveness of six advanced models for identifying financial fraud in real time: Random Forest, XGBoost, Autoencoder, Convolutional Neural Network (CNN), Weighted Deep Neural Network (DNN), and Long Short-Term Memory (LSTM) with attention. To closely replicate real-world scenarios, each model was tested using datasets with a high degree of class imbalance, a common challenge in fraud detection tasks.

Ensemble-based models, particularly Random Forest and XGBoost, achieved superior results. Both delivered outstanding accuracy scores (0.9996), F1-scores above 0.86, and Matthews Correlation Coefficients (MCC) exceeding 0.86. These figures confirm their reliable ability to detect fraudulent activity while keeping false positives to a minimum. Among the two, XGBoost slightly outperformed Random Forest in recall, suggesting its greater effectiveness at detecting fraudulent transactions—a critical quality in this domain.

Deep learning methods also proved valuable. After thorough preprocessing and the implementation of oversampling techniques, the CNN excelled in both precision and recall, showing its strength in extracting spatial features from transaction data. The LSTM with attention mechanism reached an accuracy of 87.6% on test data, demonstrating its ability to analyze temporal patterns and recognize evolving, complex fraud behaviors.

Class-weighted DNNs with regularization produced a commendable ROC-AUC of 0.91, yet their fraud-detection power varied greatly with threshold settings. This revealed an important tension between achieving higher precision and better recall. Conversely, the Autoencoder—an unsupervised method focused on anomaly detection—struggled on heavily imbalanced data, resulting in near-zero precision and recall. This suggests that relying solely on reconstruction error does not suffice for identifying fraud when legitimate transactions vastly outnumber suspicious ones.

Overall, these results indicate that while traditional ensemble algorithms are reliable and highly accurate, deep learning models contribute essential strengths in pattern extraction, especially when enhanced by attention mechanisms and data balancing strategies. The optimal approach may combine elements of both, such as hybrid models, which could leverage the complementary advantages for robust real-time fraud detection. Future work might explore ensemble stacking, adversarial training, and ongoing feedback loops for improved adaptability, as well as integrating user behavior analytics and ensuring continual updates to stay ahead of rapidly evolving fraudulent techniques.

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